KAPASA MAKASA UNIVERSITY

ENTOMOLOGY

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UNIT FIVE

TOPIC: OUTLINING INSECT ECOLOGY AND BEHAVIOUR

Objectives: by the end of this unit, SSBAT;

1. **Understand insect behaviour**
2. **Explain insect specificity**
3. **Search for host plants**
4. **Ovipositional preferences and food selection**

Introduction

Insect ecology deals with the study of the insect interactions with its habitat or environment while insect behaviour deals with the study of the "observable physiology" or the various activities that the insect displays in its habitat or environment. In order to understand the various insect behaviour, there is need to know the relationship between the plant and the insect. Therefore, information is needed on food habits, ovi-position pattern, fecundity (ability to produce young ones or to lay eggs) and effect of the environment on insect population.

Behaviour adaptation

Adaptation is any change over time that helps the insect to fit in their surroundings. Insects change shape, colour and behaviour in order to suit in a certain environment.

Camouflage

This can be termed as the ability of insects to utilize the host plants i.e. by the way insects hide themselves from the enemy. Besides hiding from the enemy, insects can generate a number of behavioural activities before they establish themselves on the host. These activities include the following:

1. Host specificity
2. Sequential activities for host selection
3. Dispersal and search for host
4. Orientation and recognition of host
5. Individual variability
6. Ovi – positional preference and food selection

1. Host specificity

Most insects are plant feeders but no plant species can be susceptible to all insect species and no insect species can be susceptible to all insect species and no insect species can feed on all plants. Insects are very specific in their feeding habits. Host specificity refers to a range of plant species on which an insect is known to occur in nature e.g. caterpillars only occur on certain tree species. Therefore, insects are divided into three categories based on host plant specificity;

a. Monophagus: Insects that are restricted to plants of a single species or may be plant specificity.
b. Oliphagus: Insects that are restricted to feed on plants of one family or members of related plant families.
c. Polyphagus: Insects that are restricted to feed on plants of more than one order.

Insects can also be divided based on the part of the plant they eat or feeds on;

i. Defoliators (leaf feeders , miners, skeletonizers)
ii. Stem borers
iii. Root feeders

2. Sequential activities for host selection

Host plant selection is a behavioural sequence by which an insect is able to distinguish a host plant from a non - host plant. There is always a behavioural pattern in which insects can select a host plant from other plants e.g. maize stalk borers. The selection of the maize stalk is the function of the female as she will deposit the eggs on the maize stalk. Usually a number of activities will take place before the selection of the host plant e.g. a hungry leaf hopper will locate their host plant by means of short range stimuli. The process may include stimulation by the colour of the plant. As it moves towards the tree, the olifactory stimuli will stimulate it. As it lands on the plant, it releases the proboscis into the plant. As it pushes its proboscis into the plant, it will look at the colour and contact stimuli. The tissue acidity stimuli will direct the insect to the desired place to settle. The customary stimuli found in the fauna will stimulate the leaf hopper for continuous feeding.

3. **Dispersal and search for host**

   Movement activities of insects is the manifestation of these two drives, which serves different functions.

   a. Dispersal: this will lead an insect to move to a more homogenous insect population and possibly invade new areas. Both adults and larva do disperse.
   b. Search behaviour: This increases the chance of insect in countering stimuli initiating behaviour pattern which may lead to ovi-position or feeding e.g. Aphids may react negatively to green foliage while they act positively to blue sky.

4. **Orientation and recognition of hosts**

   There are physical and chemical factors that are involved in guiding the female insect during ovi-position. Insect orientation behaviour is where a plant can either be rejected or accepted as a host.

   Physical factors include;

   a. Colour of plants
   b. Shape
   c. Cracks or clevises on the surface of the bark or leaf

   Chemical factors

   - Volatile chemicals (pheromones)
     Can be involved in the orientation and also help stimulate biting, stinging or probing.
   - Plant odour
     Certain plants produce smells or odour that attracts the insects (queen of the night).
   - Sensory systems
     Allow them to distinguish the host plant from the non – host plants.

5. **Individual variability**

   Individual differences in behaviour in a population change. This change may be due to conditions surrounding the individuals, insects in a population may prefer a certain type of food than when an insect is alone.

6. **Ovipositional preferences and food selection**

   Female insects will look for host plants to lay their eggs so that when the larva hatches, they feed on the host or will have enough food.